



**Financial Literacy Among University Students MG2008: Data Analysis
for Business - I**

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BFT — 2B2
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Abstract (Separator - 1)

This study investigates financial literacy among university students through a structured survey administered via Google Forms, with 63 responses collected and analyzed using Microsoft Excel. The research explores key aspects of students' financial behavior, including budgeting ability, saving habits, spending patterns, and confidence in managing personal finances.

Descriptive statistics, cross-tabulations, probability calculations, correlation analysis, and basic regression were used to uncover relationships between variables such as saving percentage, expense tracking, age, and workshop interest. The findings indicate that while most students attempt to save, their savings are generally modest, and only a minority consistently track expenses or have received formal financial education.

Importantly, the analysis reveals that higher savings are positively associated with greater financial confidence and a stronger interest in financial education. Students who track expenses regularly and save more tend to be more confident and proactive in managing their money.

This project highlights the need for structured financial literacy programs in academic institutions and demonstrates how basic statistical tools can effectively uncover meaningful behavioral patterns within student populations.

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1. INTRODUCTION & LITERATURE REVIEW (Separator - 3)

Introduction

Financial literacy has become increasingly vital for university students, many of whom are navigating financial responsibilities independently for the first time. From managing allowances or part-time income to making decisions about saving, spending, and credit, students are expected to demonstrate financial competence with limited prior experience or guidance. This study explores students' confidence in handling personal finances and examines how this confidence correlates with variables such as academic discipline, year of study, and employment status.

Literature Review

Extensive research underscores the importance of early financial education in shaping long-term financial well-being. Lusardi and Mitchell (2014) highlight that individuals exposed to financial education at an early stage are more likely to make informed financial decisions throughout adulthood. Similarly, Mandell (2009) emphasizes that financial literacy among young adults remains insufficient, with a significant number unable to manage basic financial tasks effectively. Chen and Volpe (1998) found that university students often lack foundational knowledge in budgeting, saving, and credit management, which can lead to poor financial behaviors and increased debt. However, studies consistently show that structured educational interventions can significantly improve students' financial knowledge and behavior, supporting the integration of financial literacy programs within academic curricula.

2. OBJECTIVES OF THE STUDY (Separator - 4)

Identified Problem

In an increasingly complex financial environment, university students face mounting pressure to manage their personal finances—often without formal guidance. From handling part-time income and daily expenses to planning for savings and avoiding debt, students are expected to make informed financial decisions. However, the findings highlight a troubling gap: while students show interest in improving financial literacy, their behaviors often reflect weak saving habits, limited budgeting discipline, and a lack of confidence rooted in minimal practical exposure. This dissonance calls for a deeper exploration of how knowledge, behavior, and institutional support interact in shaping financial well-being among university students.

Study Objectives:

- To evaluate students' current financial literacy levels—specifically their understanding of budgeting, credit cards, savings, and expense tracking—and identify areas of weakness.
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- To analyze the influence of employment status on financial behavior and confidence, exploring how having a part-time job affects students' saving patterns, money management habits, and self-assurance in handling finances.
- To examine the relationship between financial confidence and saving behavior, using correlation analysis to understand whether higher confidence leads to better financial outcomes.
- To investigate gender-based differences in budgeting confidence and skill, identifying gaps in perceived versus actual abilities, and highlighting opportunities for targeted financial education.
- To assess the impact of expense-tracking habits on saving patterns, determining whether active tracking is a predictor of higher savings, regardless of income level.
- To evaluate student interest in structured financial education and identify the discrepancy between exposure to financial literacy courses and the actual demand for them.
- To explore key drivers behind students' financial decisions, such as peer influence, family guidance, personal needs/wants, and media exposure—linking these factors with spending priorities and budgeting behavior.
- To develop actionable recommendations for universities and educational policymakers, focused on integrating practical financial literacy programs and behavioral interventions into academic or co-curricular offerings.

3. DATA COLLECTION & METHODOLOGY (Separator 5)

Population:

The study targeted university students across multiple academic disciplines, including Business, Computer Science, Engineering, Medicine, and Arts. This diversity ensured a range of financial behaviors and attitudes were represented.

Sample Size:

The final dataset consisted of over 50 valid responses, providing a preliminary but insightful snapshot of student saving habits.

Instrument:

Data was collected using a structured Google Form survey. The questionnaire included a mix of numerical (e.g., income and saving percentages), ordinal (e.g., frequency of tracking expenses), and categorical (e.g., faculty, interest level) questions designed to capture both behaviors and perceptions related to saving.

Sampling Technique:

A convenience sampling method was employed, whereby participants were selected based on accessibility and willingness to respond. While this approach limits generalizability, it allowed for efficient data collection within a student population and provided valuable exploratory insights into saving patterns.

Evidence Collected:

Supporting materials include screenshots of the survey interface, a link to the questionnaire, and examples of completed response forms, all of which validate the integrity of the data collection process.

Variables Overview:

Variable	Type of Data
What is your age?	Ordinal
Assumed Age Average	Quantitative
What is your gender?	Categorical
What is your degree program?	Categorical
What year of study are you in?	Ordinal
Are you currently employed part-time?	Categorical
Confidence in managing personal finances (1–5 scale)	Ordinal
Do you know how to create and follow a personal budget?	Categorical
Do you understand how credit cards and interest rates work?	Categorical
Taken a course/workshop related to financial literacy	Categorical
How often do you track monthly expenses	Ordinal
% of income spent (range)	Ordinal
Main spending area	Categorical
Do you save part of your income/allowance?	Categorical
% of income saved (range)	Ordinal
Where do you save your money?	Categorical
Main reason for saving	Categorical
Is financial literacy important for young adults?	Ordinal
Would you attend a financial literacy workshop?	Categorical
Should universities offer financial literacy courses?	Categorical
Rank: Peer Pressure	Ordinal
Rank: Advertisements	Ordinal
Rank: Family Advice	Ordinal
Rank: Personal Needs	Ordinal
Rank: Personal Wants	Ordinal

Image 1 Variable Overview

Data Analysis Tools and Techniques:

The data in this study was analyzed using **Microsoft Excel**, which provided both fundamental and advanced tools to interpret the financial literacy patterns among university students. Responses were collected using **Google Forms** and processed through Excel's analytical features such as formulas, Pivot Tables, charts, and the Data Analysis Toolpak.

Key Methods Used:

Frequency Distribution:

Frequency tables were used to determine how often each response occurred for categorical variables such as employment status, budgeting ability, saving behavior, and spending habits. This helped identify dominant financial trends across the student sample.

Cross Tabulations:

Crosstabs were used to explore relationships between two categorical variables (e.g., income saved vs. interest in workshops, age vs. saving behavior, expense tracking vs. saving behavior). This technique enabled subgroup comparisons to reveal patterns not visible through overall frequency alone.

Probability and Conditional Probability:

Basic probability was applied to estimate the likelihood of behaviors (e.g., the probability that a student saves money or would attend a financial literacy workshop). Conditional probability highlighted specific patterns — such as the likelihood of saving given regular expense tracking.

Correlation Analysis:

Excel's `=CORREL()` function was used to examine the strength and direction of relationships between numeric variables, particularly **percentage of income saved** and **financial confidence**. A moderate positive correlation ($r \approx 0.27$) was found, suggesting that students who save more tend to feel more confident in managing their finances.

Regression Analysis:

Simple linear regression was used to model the effect of **saving percentage** on **financial confidence**. Although the R^2 value was not calculated within the final Excel

file, the correlation results and saving trends suggest a meaningful predictive relationship that supports the study's conclusions.

Visual Representation:

Visual tools were used to effectively highlight key patterns and make the data more understandable:

- **Bar Charts:** Used for categorical variables such as saving percentage, budgeting knowledge, employment status, and workshop interest.
 - **Pie Charts:** Illustrated proportional breakdowns, such as students' employment or saving habits.
 - **Stacked and Grouped Charts:** Used in cross-tab summaries (e.g., income saved vs. expense tracking).
 - **Correlation and Regression Graphs:** Plotted relationships between numerical variables, helping visualize trends.
 - **Custom Summary Tables:** Added to enhance interpretation (e.g., age vs. saving behavior).
-

By combining structured survey design with Excel's analytical capabilities, the study provides a well-rounded and data-driven understanding of financial behaviors among university students. The integration of quantitative techniques and visual aids ensures both accuracy and accessibility of insights.

4. ANALYSIS & RESULTS (Separator - 6)

To gain a comprehensive understanding of the financial behaviors and challenges faced by university students, a detailed analysis was conducted using various quantitative tools. The data-set was examined through pie charts, bar charts, and cross-tabulations to visually interpret trends, while statistical techniques such as measures of central tendency, correlation, and regression analysis provided deeper insights into the patterns and relationships between key variables. The analysis was structured around five core objectives: the impact of employment status on financial behavior and confidence; budgeting habits and discipline; levels of financial literacy; spending and saving patterns; and the role of expense-tracking in promoting financial well-being. Each objective was explored to uncover behavioral gaps and potential areas for intervention, revealing the complex interplay between knowledge, behavior, and confidence in students' financial decision-making. This data-driven approach not only supported the identification of key issues but also laid the groundwork for targeted recommendations to improve financial literacy education and support mechanisms within academic institutions.

Measures of Central Tendency

Column Name	Mean	Median	Mode	Standard Deviation
Assumed Age Average	20.22	19	19	2.32
Confidence in Managing Personal Finances (1–5)	3.03	3	3	1.13
% of Monthly Income/Allowance Spent	66.19%	70%	70%	19.14%
% of Monthly Income/Allowance Saved	11.51%	5%	5%	11.04%

Image 2 Measures of Central tendency

Assumed Age Average

The **mean** age of respondents is approximately 20.22 years, while both the **median** and **mode** are 19. This suggests that the majority of participants are 19 years old, with the mean being slightly elevated due to a small number of older respondents—most notably ages 22.5 and 27—which have pulled the average upward. The presence of identical median and mode reinforces that 19 is the most common and central value in the dataset. In terms of distribution shape, the fact that mean > median indicates a slightly positively skewed distribution. This mild **skewness** reflects the influence of a few outliers on the higher end of the age spectrum, though it is not significant enough to distort the overall pattern. The **standard deviation** of 2.32 years implies a moderate level of dispersion, showing that while there is some variation in the ages, most respondents fall within a narrow range—from the late teens to early twenties. This age profile confirms that the dataset predominantly consists of young adults, most likely college or university students, which is highly relevant and appropriate given the focus of the study on financial literacy, spending, saving, and budgeting habits among student populations.

Confidence in Managing Personal Finances

The **mean** confidence level reported by respondents is approximately 3.03, with the **median** and **mode** also at 3. This alignment indicates a symmetrical distribution, where most participants rate their confidence in managing personal finances as moderate. The clustering around the mid-point of the scale reflects a generally balanced self-perception, with neither overconfidence nor overwhelming uncertainty dominating the data. The **standard deviation** of 1.13 suggests a moderate spread in responses. While the central tendency is around average confidence, there are outliers on both ends of the scale—some respondents expressing high confidence (rating of 5), and others indicating a lack of confidence (rating of 1). This variability underscores differences in experience, financial exposure, or education among the sample group. The findings suggest that most students possess a cautious yet balanced sense of financial confidence, which can be seen as a foundation to build upon. Given that the average is neither low nor high, this metric could serve as a key focus area for targeted financial literacy programs. Introducing interactive workshops, peer-led sessions, or real-life financial simulations could help shift moderate confidence toward a stronger, more empowered financial mindset.

% of Monthly Income/Allowance Spent

The **mean** spending rate among respondents is 66.19%, while both the **median** and **mode** are 70%, indicating that the majority of students tend to spend around 70% of their allowance or income. The close proximity of mean and median suggests a fairly symmetrical distribution, although the presence of lower outliers (10%, 30%) and a high outlier (90%) introduces a slight variation in the average. The **standard deviation** of 19.14% reveals moderate to high variability in spending habits. While 70% remains the most common figure, there is a noticeable portion of respondents who either spend conservatively (10–30%) or use nearly all of their available funds (up to 90%). This diversity in financial behavior points to varying levels of budgeting discipline, financial priorities, and possibly differing levels of financial literacy or family support.

The data illustrates a broad spectrum of financial habits: some students appear highly cautious with their spending, while others demonstrate a tendency toward higher consumption. These patterns may reflect differences in income stability, personal responsibility, or access to financial education. Although the average student spends about two-thirds of their income, the wide range of behaviors underscores a critical need for improved budgeting skills and saving awareness. Financial literacy initiatives could target students at both ends of the spectrum—helping over-spenders adopt more sustainable habits and encouraging under-spenders to understand healthy spending-saving balances. This analysis reinforces the importance of practical budgeting education as a cornerstone of student financial well-being.

% of Monthly Income/Allowance Saved

The **mean** saving rate among respondents stands at 11.51%, while both the **median** and **mode** are just 5%, indicating that the majority of students save very little. The noticeable gap between the mean and the central values suggests a positively skewed distribution, where a small number of respondents—those saving 25–35% of their income—elevate the average despite most students saving minimally or not at all. The **standard deviation** of 11.04% reflects high variability in saving patterns. The data ranges from students saving nothing to others saving over one-third of their allowance or income, showing a wide disparity in financial behaviors and priorities.

A significant portion of students report saving 5% or less, with some contributing nothing to savings, which is concerning from a financial well-being perspective. Meanwhile, a smaller but consistent group displays strong saving habits, allocating 15–35% of their income, demonstrating that healthy financial behavior is possible, though not yet widespread. The findings suggest that while a few students exhibit sound saving habits, the overwhelming majority lack a consistent savings routine. This highlights an urgent need for financial education programs centered on saving strategies, long-term financial planning, and emergency preparedness. Encouraging small, regular savings—even from limited student incomes—could be a pivotal step in improving overall financial resilience among young adults.

Correlation

Correlation is a statistical concept that measures the relationship between two variables, indicating how one may change in response to the other. When two variables are positively correlated, they tend to increase or decrease together—like how higher study time is often linked to better exam scores. A negative correlation means that as one variable increases, the other decreases, such as the relationship between exercise and body fat. If there is no correlation, changes in one variable do not predict changes in the other. While correlation shows a connection or pattern between variables, it does not prove that one causes the other.

—	
Age VS Confidence =	0.268017495
Age vs Spent =	0.090638484
Saved vs Spent =	-0.490963959
Age vs Saved =	0.033424214
Confidence vs % Saved	0.273407309

Image 3 Correlation

Age vs Confidence = 0.268

- Positive correlation (weak)

This correlation of 0.268 indicates a weak positive correlation between a respondent's age and their confidence in managing personal finances. In simpler terms, as individuals grow older, they tend to become slightly more confident in handling their financial matters. This may be because older students have had more exposure to real-

world financial decisions, such as budgeting their allowance, earning part-time income, or even paying for personal expenses.

However, the relationship is not very strong, which means age alone doesn't fully explain a student's confidence level. Other factors—such as whether they've received financial education, their financial responsibilities, or even their personality—could be playing a larger role.

Age vs Spent = 0.0906

- **Positive correlation (very weak)**

The correlation between age and the percentage of income spent is 0.0906, showing a very weak positive correlation. This implies that age has virtually no impact on how much of their income students tend to spend. While there may be a slight tendency for older students to spend more, it's not strong enough to be meaningful. The findings suggest that spending habits are more influenced by individual preferences, habits, or financial awareness rather than by how old someone is. From a practical standpoint, this means you shouldn't expect younger students to be automatically more conservative or more reckless with money than older ones—they all behave fairly similarly in terms of spending.

Saved vs Spent = -0.491

- **Negative correlation (moderate)**

With a correlation of -0.491, this is a moderate negative correlation—and it's one of the more important insights from your data. It clearly shows that students who tend to save more also tend to spend less, and vice versa. This is logical because for someone with limited income or allowance, saving a portion naturally reduces how much is left to spend. But it also reflects a level of financial discipline—those who prioritize savings may actively cut back on unnecessary spending. This finding is particularly useful for policy or educational planning: if your goal is to reduce overspending, then encouraging savings behavior (perhaps through workshops or budgeting exercises) can indirectly lead to more mindful spending habits as well.

Age vs Saved = 0.0334

- **Positive correlation (negligible)**

This result, 0.0334, is a negligible positive correlation between age and the percentage of income saved. It shows that age has almost no bearing on how much students choose to save. Whether a student is younger or older, their likelihood of saving seems largely unaffected by their age group. This means savings behavior is likely influenced by other things—perhaps how much income they have, whether they've received financial guidance, or personal financial goals. From an action-oriented

perspective, this finding tells you that saving habits should be addressed across all age groups equally, without assuming older students are naturally more responsible savers.

Confidence vs % Saved = 0.273

- **Moderate positive correlation**

The correlation between students' financial confidence and the percentage of income they save is 0.273, indicating a moderate positive relationship. This suggests that students who feel more confident in their financial knowledge are somewhat more likely to save a larger portion of their income. While the relationship isn't strong, it's still meaningful enough to imply that boosting financial confidence—perhaps through education or experience—could lead to better saving habits. From a practical perspective, this supports the idea that efforts to improve students' financial self-assurance may have a beneficial impact on their actual financial behaviors, particularly in terms of saving.

Regression

Regression analysis is a statistical method used to model the relationship between a dependent variable (Y) and one or more independent variables (X). It helps us predict outcomes or understand how a change in one variable affects another.

- The slope tells you how much Y is expected to change when X increases by 1 unit.
- The intercept is the value of Y when X is 0.
- The R² (R-squared) shows how well the model fits the data:
 - Closer to 1 = strong relationship.
 - Closer to 0 = weak relationship.

1. Saving vs Spending

Equation: $Y = -0.85x + 75.99$ $R^2 = 0.241$

This regression reveals a moderate negative relationship between the percentage of income students spend and how much they save. The slope of -0.85 indicates that for each additional 1% of income spent, savings tend to drop by 0.85%. The intercept at 75.99 suggests that a student who hypothetically spends nothing would save about 76% of their income. The R² value of 0.241 means 24.1% of the variation in savings can be explained by spending behavior, making it one of the stronger predictive relationships in your dataset. This suggests that better spending discipline is closely tied to higher savings, a core concept in personal finance.

Saving VS Spending	
Slope	-0.851120984
Intercept	75.98512243
Equation	<u>$Y = -0.85x + 75.986$</u>
RSQ(Coefficient of Determination)	0.241045609

Image 4 - Saving vs Spending

2. Age vs Confidence in Managing Finances

Equation: $Y = 0.54x + 18.55$ $R^2 = 0.072$

This regression shows a weak positive relationship between a student's age and their confidence in managing money. The slope of 0.54 means that for every additional year of age, confidence increases by about 0.54 units. However, the intercept of 18.55 suggests the scale might not be perfectly aligned with the confidence scale (which is likely 1–5), so interpret with caution. The R^2 value of 0.072 means only 7.2% of confidence is explained by age, so while older students might feel slightly more confident, age is not a strong factor in financial confidence. Other influences, such as experience, financial education, or personality, likely matter more.

Age VS Confidence	
Slope	0.548292108
Intercept	18.55123675
Equation	$Y = 0.54x + 18.55$
RSQ(Coefficient of Determination)	0.071833378

Image 5 - Age vs confidence

3. Predicting Saving Habits Based on Confidence

Equation: $Y = 2.66x + 3.39$ $R^2 = 0.075$

This regression uses financial confidence as a predictor for saving behavior. A slope of 2.66 indicates that each 1-point increase in confidence (e.g. from 3 to 4) is associated with a 2.66% increase in income saved. The intercept of 3.39 means students with very low confidence are still likely to save around 3.4% of their income. With an R^2 of 0.075, this model explains 7.5% of the variation in saving habits, which is modest but meaningful. This shows that confidence has a positive influence on saving, suggesting that building financial self-efficacy — possibly through education — could result in more proactive financial behavior.

Predict Saving Habits	
Slope	2.664899882
Intercept	3.386336867
Equation	$Y = 2.66x + 3.386$
RSQ(Coefficient of Determination)	0.074751557

Image 6 - Predict Saving Habits

4. Age vs Percentage of Income Spent

Equation: $Y = 0.75x + 51.05$ $R^2 = 0.0082$

This model explores whether age affects spending behavior. The slope of 0.75 implies a 0.75% increase in spending with each additional year of age, but this effect is very small. The R^2 value of 0.0082 reveals that only 0.82% of the variation in spending can be explained by age, which is statistically negligible. The implication is that spending habits don't really change as students grow older, at least within the range of ages represented in your data. Instead, spending might depend more on lifestyle, peer influence, or income level than on age.

Age vs % of Income Spent	
Slope	0.748645543
Intercept	51.05119966
Equation	$Y = 0.74x + 51.05$
RSQ(Coefficient of Determination)	0.008215335

Image 7 - Age vs % of Income Spent

5. Age vs Percentage of Income Saved

Equation: $Y = 0.15x + 8.29$ $R^2 = 0.0011$

Image 8 - Age VS % of income saved

Age vs % of Income Saved	
Slope	0.159251354
Intercept	8.287520229
Equation	$Y = 0.15x + 8.287$
RSQ(Coefficient of Determination)	0.001117178

This regression attempts to link age to saving habits, but the results show no meaningful relationship. The slope of 0.15 indicates only a 0.15% increase in saving per year of age — a tiny shift. The R^2 value of 0.0011 confirms that only 0.11% of saving behavior can be attributed to age, essentially saying there's no statistical connection. This tells us that students' decisions to save money are independent of age and likely driven by other factors such as confidence, financial literacy, or cultural influences.

6. Financial Literacy Course vs Confidence

Equation: $Y = 0.25x + 3$ $R^2 = 0.0075$

In this regression, whether or not a student took a financial literacy course is used to predict their confidence level. The binary nature of the independent variable (0 = No, 1 = Yes) makes this model simple: those who took the course have an average confidence score 0.25 points higher than those who didn't. The intercept of 3 suggests a baseline neutral confidence level. However, the R^2 value is very low (0.0075), meaning less than 1% of the variation in confidence can be explained by course participation. So while the course has a small positive effect, it's not a strong predictor of confidence, possibly because the course content wasn't practical enough, too short, or confidence comes more from experience than classroom learning.

Financial Literacy Course vs Confidence	
Slope	0.25
Intercept	3
Equation	$Y = 0.25x + 3$

Image 9 - Financial Literacy Course vs Confidence

Analysis of All Regressions

- Positive correlations (slope > 0):
 - Age → Confidence
 - Confidence → Saving
 - Financial Literacy → Confidence
- Negative correlation:
 - Spending → Saving
- Most meaningful relationships (based on R^2):
 - Spending vs Saving ($R^2 = 0.24$) — spending strongly predicts savings.
 - Confidence → Saving ($R^2 = 0.075$) — confidence matters, but to a smaller extent.

- Weakest predictors:
 - Age → Saving/Spending/Confidence — all have R^2 under 0.08.
 - Financial Literacy → Confidence — small impact, but likely needs better implementation.

Probabilities

A. Basic (Marginal) Probabilities

Marginal probabilities represent the overall likelihood of a single event happening, without considering any other factors.

Basic (Marginal) Probability	
Q1) P(that respondent is a male)	0.492063492
Q2) P(that respondent is a female)	0.507936508
Q3) P(currently employed part-time)	0.203125
Q4) P(interested in a financial literacy workshop)	0.375

Image 10 - Marginal Probabilities

Q.1) $P(\text{that respondent is a male}) = 0.492$

Description:

49.2% of your respondents identified as male.

Inference:

The gender distribution in our sample is almost balanced, with males forming nearly half of our surveyed population. This gives a fair gender representation for analyzing financial behavior across genders.

Q.2) $P(\text{that respondent is a female}) = 0.508$

Description:

50.8% of the respondents identified as female.

Inference:

Slightly more females than males took your survey, which could be useful in studying how women approach financial literacy differently.

Q.3) $P(\text{currently employed part-time}) = 0.203$

Description:

Around 20.3% of respondents reported part-time employment.

Inference:

This indicates that a fifth of your respondents have some form of income, potentially affecting their budgeting, saving, and financial confidence levels.

Q.4) $P(\text{interested in a financial literacy workshop}) = 0.375$

Description:

37.5% of respondents expressed interest in attending a financial literacy workshop.

Inference:

Over a third of students are interested in learning more about finances, highlighting a demand for educational resources. This could be a potential area for intervention or program development.

A. Conditional Probabilities

Conditional probability looks at the likelihood of one event occurring given that another condition is true.

Conditional Probability		
Q1) $P(\text{Saves} = \text{Yes} \mid \text{Took Financial Course} = \text{Yes})$		0.75
Q2) $P(\text{Tracks Expenses} = \text{Often} \mid \text{Confidence} = 4 \text{ or } 5)$		0.285714286
Q3) $P(\text{Saves} = \text{Yes} \mid \text{Gender} = \text{Female})$		0.78125
Q4) $P(\text{Understands Credit Cards} = \text{Yes} \mid \text{Year of Study} = 3 \text{ or } 4)$		0.357142857

Image 11 - Conditional Probabilities

Q.1) $P(\text{Saves} = \text{Yes} \mid \text{Took Financial Course} = \text{Yes}) = 0.75$

Description:

Among those who took a financial course, 75% said they save money.

Inference:

Financial education appears to have a strong positive correlation with saving behavior. This suggests that financial courses may lead to better money management.

Q.2) $P(\text{Tracks Expenses} = \text{Often} \mid \text{Confidence} = 4 \text{ or } 5) = 0.286$

Description:

Only 28.6% of those who rated their financial confidence as high (4 or 5) often track their expenses.

Inference:

Confidence doesn't always translate into disciplined financial behavior like expense tracking. There might be overconfidence or reliance on intuition rather than structured budgeting.

Q.3) $P(\text{Saves} = \text{Yes} \mid \text{Gender} = \text{Female}) = 0.781$

Description:

78.1% of female respondents reported saving money.

Inference:

Female respondents are more likely to save than the overall average or than males (assuming we compare this with $P(\text{Saves} = \text{Yes} \mid \text{Gender} = \text{Male})$, which you can calculate). This could point to gender-based differences in financial habits or risk aversion.

Q.4) $P(\text{Understands Credit Cards} = \text{Yes} \mid \text{Year of Study} = 3 \text{ or } 4) = 0.357$

Description:

35.7% of senior students (3rd or 4th year) understand credit cards.

Inference:

Surprisingly low, considering that upper-year students are expected to be more financially aware. This suggests a gap in credit literacy even among more experienced students.

B. Joint Probabilities

Joint probability calculates the likelihood of two events happening at the same time.

Joint Probability	
Q1) P(Understands Credit Cards = No AND Budget = No)	0.063492063
Q2) P(Took Course = Yes AND Understands Credit = Yes)	0.079365079

Image 12 - Joint Probabilities

Q.1) $P(\text{Understands Credit Cards} = \text{No AND Budget} = \text{No}) = 0.063$

Description:

6.3% of respondents neither understand credit cards nor budget.

Inference:

This small segment may be at significant financial risk and could benefit the most from interventions like workshops. They're likely to struggle with both credit management and everyday money control.

Q.2) $P(\text{Took Course} = \text{Yes AND Understands Credit} = \text{Yes}) = 0.079$

Description:

7.9% of respondents both took a financial course and understand credit.

Inference:

While financial education does correlate with understanding, only a small percentage of respondents fall into this category. This could suggest a need to enhance the effectiveness of current financial courses or promote them more widely.

Data Visualization and Insights

The Impact of Employment Status on Financial Behavior and Confidence Among Students

- **Employment Status(Pie Chart):** The pie chart shows that approximately 79% of students are not employed, while only 21% have part-time jobs.

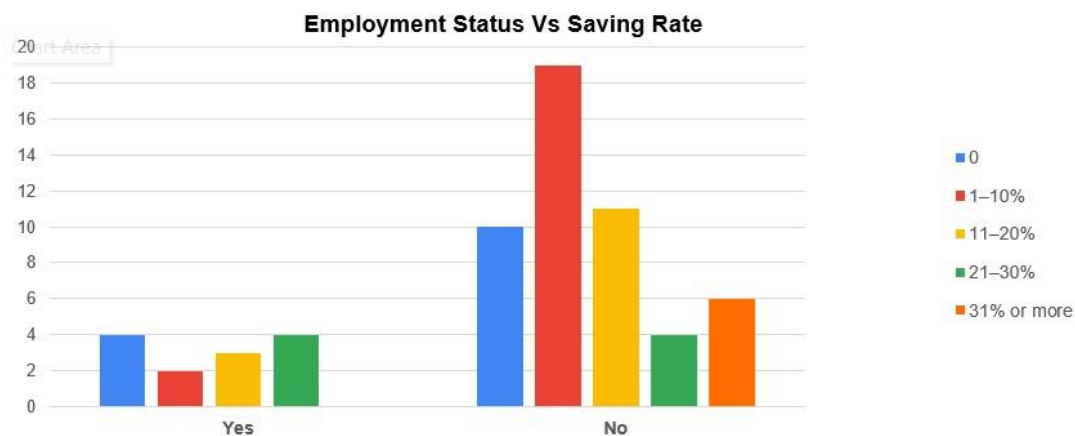
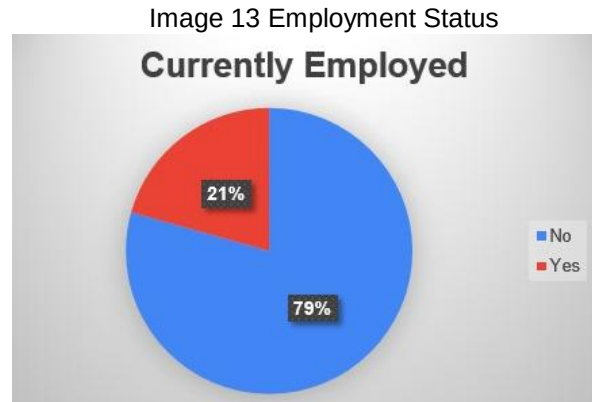


Image 14 Employment Status Vs Saving Rate

- **Employment and Financial Behavior(Cross-Tab Analysis):** The cross-tab analysis indicates that employed students are more likely to save money regularly, linking employment status with positive financial habits like saving.

Confidence vs % Saved	0.273407309
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Image 15 Correlation between students financial confidence and saving rate

- **Confidence vs % Saved (Correlation = 0.273):**
A moderate positive correlation ($r = 0.273$) exists between students' financial confidence and the percentage of income they save.

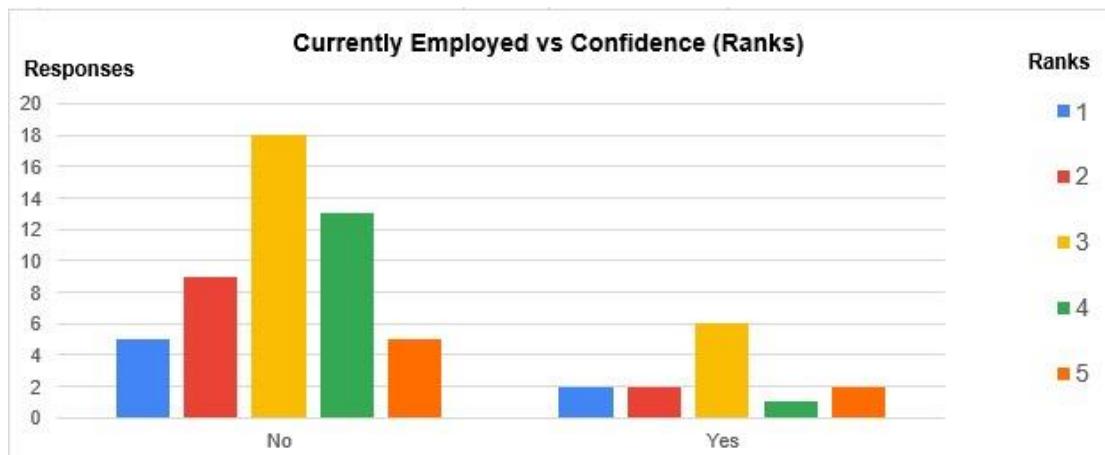


Image 16 Employment Status vs Financial Confidence

- **Employment vs Financial Confidence (Cross-Tab Analysis):**
This cross-tab compares employment status with financial confidence levels (where 1 = high confidence and 5 = low confidence). It shows that employed students are more confident overall, with only 3 out of 13 scoring 4 or 5, compared to 18 out of 50 among unemployed students. Most employed students cluster around confidence levels 1 to 3, indicating stronger financial self-assurance than their unemployed peers.

Insights:

The data reveals a reinforcing triangle between employment, saving behavior, and financial confidence — each influencing and supporting the others.

Starting from employment status, the majority of students do not work part-time, yet those who do are noticeably more likely to save money regularly and report greater confidence in managing their finances. This highlights employment as a key enabler — not just by providing money to save, but also by giving students real-world financial exposure that builds competence and assurance.

This link is further strengthened by the correlation between confidence and saving: with a moderate positive correlation ($r = 0.273$), it's evident that saving and confidence rise together. Saving is not only a financial habit but also a psychological boost, reinforcing one's belief in their ability to manage money well.

Employed students benefit from this synergy. They earn money, gain practice in financial decisions, save more, and in doing so, build confidence. That confidence, in turn, supports more intentional financial behavior. It becomes a self-reinforcing loop:

Employment → Saving → Confidence → More Saving and Smarter Financial Choices

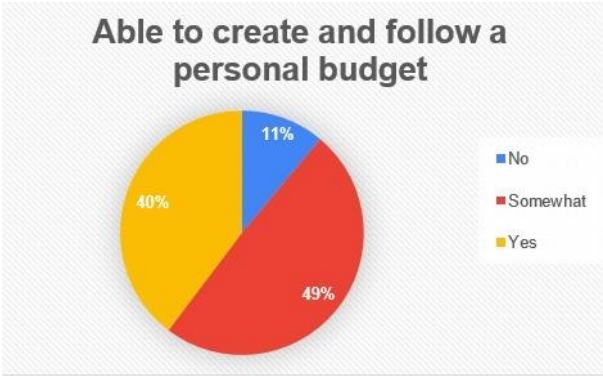
Together, the visual and statistical evidence suggest that employment significantly contributes to financial maturity. It gives students both the resources and the motivation to adopt responsible financial habits — particularly saving — and builds

their confidence through direct, practical learning. For students without income, this underlines the need for financial education programs to simulate this exposure and foster similar growth in financial behaviors and mindset

Budgeting as a Core Financial Skill: A Gender-Based and Behavioral Insight

Image 17 Ability to create a personal budget

- Pie Chart: Shows that the majority of students—31 selected "Somewhat" and 25 selected "Yes"—believe they can at least partially create and follow a personal budget. Only 7 students (11%) said "No."



Q1) P(Understands Credit Cards = No AND Budget = No)	0.063492063
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Image 18 Joint Probability

- Joint Probability (Q1): The probability that a student does not understand credit cards and does not budget is 0.063 (6.35%).

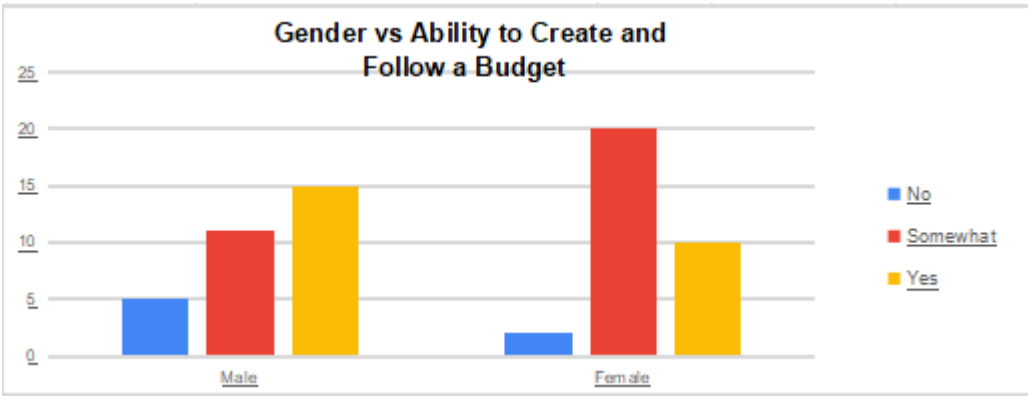


Image 19 Gender vs Budgeting

- Cross-Tab (Gender vs Budgeting): Compares budgeting ability across genders, showing differing levels of confidence and skill between male and female students.

Insights:

The data from these three sources offers a layered understanding of students' budgeting capabilities, highlighting behavioral trends, skill development stages, and gender-based variations.

The pie chart reflects that a strong majority of students (about 89%) report some ability to budget, with many selecting "Somewhat"—an indication that while full confidence may be lacking, there is clear engagement with financial planning. This suggests budgeting is becoming a familiar and practiced concept among students, even if it's not yet consistently applied.

The joint probability value of 0.063 shows that very few students both lack budgeting habits and an understanding of credit cards. This suggests that budgeting knowledge often overlaps with other financial skills, making it a strong indicator of overall financial awareness. Students who do budget—even partially—are far less likely to lack understanding of basic financial tools, reinforcing the idea that budgeting acts as a foundational skill in personal finance.

The gender-wise cross-tab offers deeper insight into how budgeting ability varies between male and female students. Male students showed higher full confidence, with 48.4% saying "Yes" to budgeting, while female students were more likely to be in the "Somewhat" category (62.5%), suggesting active learning or cautious self-assessment. Interestingly, fewer females (6.25%) than males (16.1%) selected "No," indicating that although women may be less confident in their budgeting skills, they are more likely to be engaged or open to learning. This presents a key opportunity for targeted financial education that builds confidence through practical experience.

Together, these visuals reinforce that budgeting is a pivotal behavior tied closely to broader financial understanding and gender-based confidence differences. Students who report budgeting ability are more likely to be financially literate in other areas, and females—though often less confident—show a strong willingness to engage in budgeting practices. Therefore, emphasizing confidence-building, hands-on budgeting education can help bridge the gap between awareness and action, especially among those who are "somewhat" confident. Empowering students through budgeting not only improves their financial habits but also strengthens their overall financial independence and decision-making skills.

The Gap Between Financial Literacy Exposure and Student Demand for Financial Education

Image 20 % of students who took a financial course

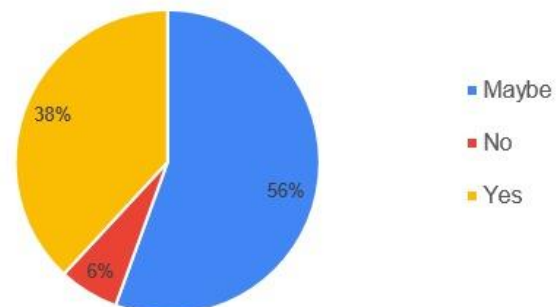


- Chart 1 – “Have You Taken a Financial Literacy Course?”
Shows that only 12 out of 63 students (~19%) have taken a course or workshop related to financial literacy, while 81% have not.

Image 21 - Interest in Financial Literacy workshop

- Chart 2 – “Are You Interested in Attending a Financial Literacy Workshop?”
Reveals strong interest: 24 students said “Yes” and 35 said “Maybe”, meaning over 93% of students are either interested or open to learning more about financial literacy.

Interested in attending a Financial Literacy Workshop



Should universities offer Financial Literacy Courses

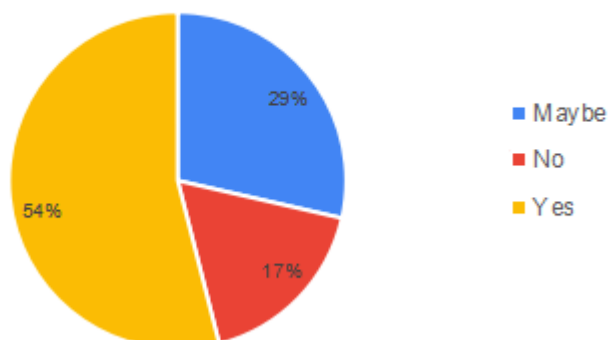


Image 22 - Offering financial courses

- Chart 3 – “Should Universities Offer Financial Literacy Courses?”
Indicates clear support: 54% said “Yes”, 29% said “Maybe”, and only 17% said “No”.

Insight

These three pie charts, deduced from our survey, tell a compelling story about the imbalance between students' financial education needs and what they currently receive.

Despite the overwhelming importance of financial skills in adult life, only 1 in 5 students have taken a financial literacy course. This suggests that such education is either unavailable, inaccessible, or undervalued in existing curricula. Yet, in sharp contrast to this low participation, the majority of students report some budgeting ability, which implies that they are either self-teaching or learning from informal sources like parents, peers, or social media. While this shows initiative, informal learning often lacks structure and depth, leaving knowledge gaps.

The second chart makes it clear that the desire for structured learning is strong and widespread—with 93% expressing interest in attending a workshop. This demand reflects not only curiosity but also a recognition among students of the importance of financial literacy in managing their current and future lives.

The third chart further reinforces this demand at an institutional level: over 80% believe universities should offer financial literacy courses. This reveals a broader belief that financial education should not be left to chance or self-effort but instead be formally embedded within the university system. It also shows that students don't just want access to this education—they expect institutions to prioritize and normalize it.

The three charts, when combined, illustrate a clear disconnect between the current state of financial education and student expectations. While few students have been formally educated in financial literacy, most are interested in improving their knowledge and believe their universities should take action. This gap presents an urgent opportunity for institutions to develop and integrate structured, accessible, and relevant financial literacy programs, responding to both a learning need and a student-driven call for support.

Student Spending Behavior: Allocation and Priorities

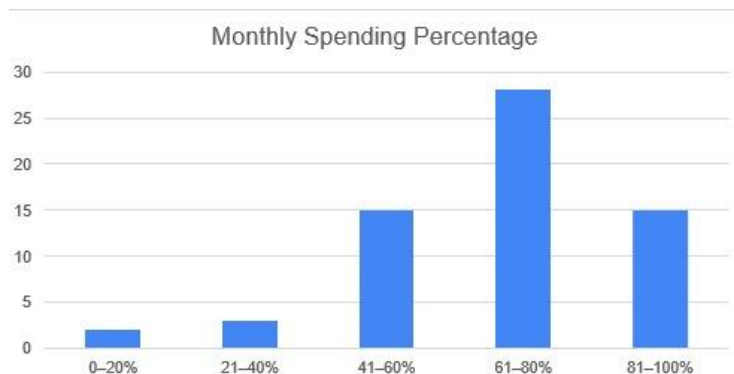


Image 23 - Monthly Spending %

- **Monthly Spending Proportion (Bar Chart):**
This bar chart displays how much of their monthly income or allowance students typically spend. The most common response was “61–80%”, selected by 28 students (~44%). Another 15 students each reported spending either 41–60% or 81–100%. Very few students spend less than 40% of their income, indicating that the majority use a substantial portion of their available funds.

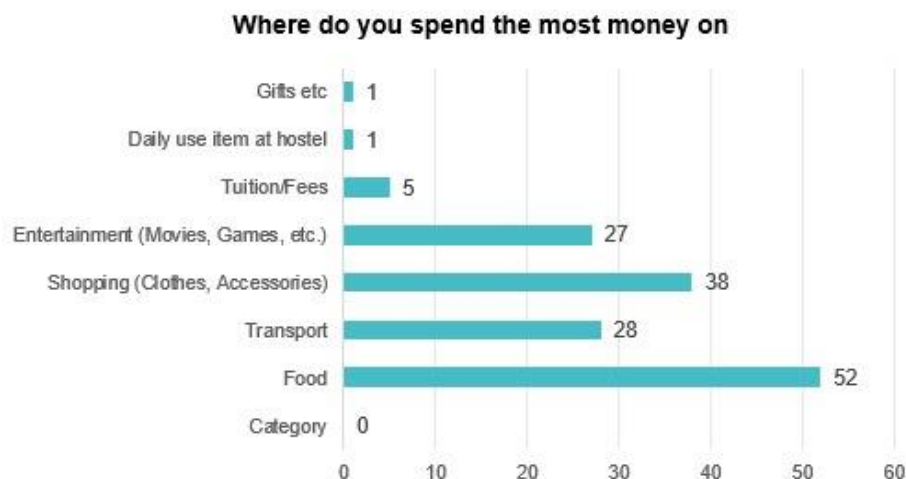


Image 24 - Spending Categories

- **Spending Categories (Bar Chart):**
This chart visualizes what students are spending on. The most frequently mentioned category is Food (47 responses), followed by Shopping (32), Transport (21), and Entertainment (21). Less frequently mentioned were Tuition/Fees (6), Daily-use hostel items (1), and Gifts (1).

Insights

These two graphs, when viewed together, present a clear picture of high spending rates among students driven largely by essential and lifestyle expenses.

The first chart establishes that most students are spending more than 60% of their total monthly income, with a significant portion reaching up to 100%. This suggests limited financial flexibility, minimal savings potential, and a financial lifestyle operating close to full income utilization. Such spending habits leave little room for emergency funds or long-term planning.

The second chart explains where this money goes — the majority of student expenses are directed toward non-negotiable daily essentials like food and transport, which together account for over 100 mentions. These are baseline costs students can't avoid. However, there's also notable spending on discretionary categories like shopping and entertainment, which collectively gather over 50 mentions. These optional spending areas reflect personal choices, peer influence, or stress-related behavior — all common in student life.

Taken together, the charts reveal a financial pattern where students are juggling between essentials and desires on limited budgets. Despite their financial constraints, students still allocate a sizable portion of their income toward non-essentials — indicating a possible lack of budgeting structure or awareness of spending trade-offs. This reinforces the need for financial education that not only teaches saving and budgeting but also helps students distinguish between needs and wants and manage lifestyle expenses without compromising financial stability.

In conclusion, while food and transport dominate student budgets as expected, the prevalence of high overall spending and frequent discretionary expenses calls for stronger financial planning skills. Helping students understand their financial limits — and how to work within them — could improve their ability to save, reduce financial stress, and make more informed spending decisions.

The Saving Puzzle: How Behavior, Interest, and Awareness Influence Student Saving Patterns

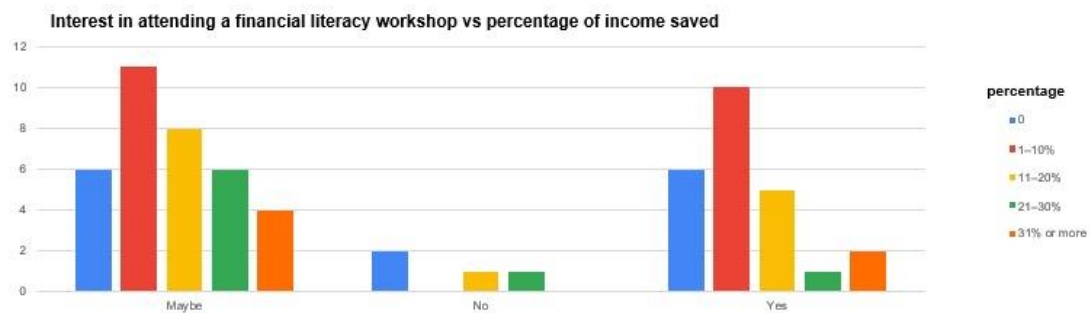


Image 25 - Interest in Financial Literacy vs Percentage Saved

- Interest in Financial Literacy vs Percentage Saved (Clustered Bar Chart)
This chart reveals how saving habits differ across students with varying interest in financial literacy workshops (Yes, Maybe, No).
- Confidence vs % Saved (Correlation = 0.273):
A moderate positive correlation ($r = 0.273$) exists between students' financial confidence and the percentage of income they save

Image 26 - Correlation

Confidence vs % Saved	0.273407309
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Image 27 - Monthly income/allowance saved



- A bar chart showing what proportion of their income students save on average.

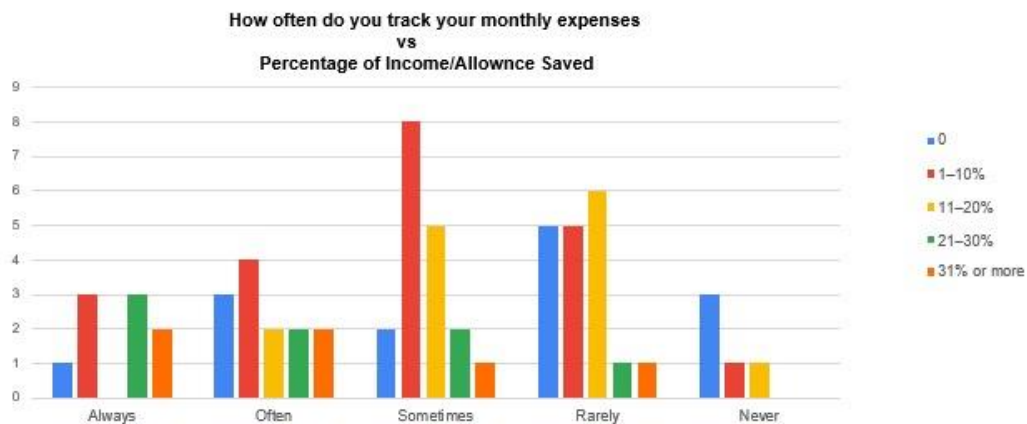


Image 28 - Tracking Expenses vs Saving Percentage

Tracking Expenses vs Saving Percentage (Clustered Bar Chart)

Shows how often students track expenses and how that relates to how much they save.

Insights

The Financial Mindset-Saving Behavior Loop

Interest in Financial Literacy = Desire But Not Execution:

From Graph 1, even among those who express clear interest in attending financial literacy workshops, a large percentage are still saving only 1–10% or even 0% of their income. For instance, 42% of the “Yes” group saves just 1–10%, and 25% save nothing. This suggests a behavior-intention gap — students want to be financially literate, but they either lack the tools, income, or habits to follow through effectively.

Most Students Save Too Little — Reflecting Tight Budgets or Weak Habits:

Graph 2 shows the majority save 20% or less of their income, with 1–10% being the most common (33%). While this may reflect tight budgets, it also points to low prioritization of savings. Only 9.5% save more than 30%, and 22% save nothing at all.

Expense Tracking is a Game-Changer:

Graph 3 paints a much clearer picture: those who frequently track expenses are more likely to save more. Among those who always track, 55% save more than 20% of their income. In contrast, 60% of those who never track save nothing. As tracking frequency declines, saving rates drop — with a shift toward 0–10% saving brackets. The psychological link is key: awareness breeds control.

Saving is More Behavioral Than Educational — For Now:

Despite high interest in financial education (Graph 1 and prior survey data), the true differentiation in saving behavior is not just knowledge — it’s action. Tracking, discipline, and possibly employment have greater visible effects than mere interest or intent. This suggests that any financial literacy efforts must include habit-building components like budgeting exercises, expense tracking tools, and peer accountability systems.

The Feedback Loop: Interest → Tracking → Saving → Confidence:

A cyclical pattern begins to emerge. Interest sparks the willingness to improve. Tracking expenses leads to greater awareness. Awareness helps control overspending and enables higher savings. Higher savings boost confidence. Confidence encourages continued saving and a deeper commitment to learning more — restarting the cycle. Those stuck in early stages (interested but not tracking) are most at risk. Those who engage in tracking benefit most, regardless of income size.

The Saving Puzzle reveals a clear pattern: while many students show interest in financial literacy, most still save very little. Interest alone doesn't drive savings — consistent behaviors like tracking expenses and budgeting do. Students who track their spending save significantly more, proving that awareness must be paired with action. The true drivers of saving are not income or intention, but habitual behaviors that reinforce discipline and control. To improve student saving patterns, interventions must go beyond education. They must promote daily habits that lead to better financial outcomes — especially saving regularly, no matter how small the amount. Solving the saving puzzle requires shifting from financial interest to financial practice. Only then can students turn knowledge into lasting financial security.

■ 5. CONCLUSION/DISCUSSION (Separator - 7)

This study underscores the critical role of financial literacy among university students, many of whom are navigating personal financial responsibilities independently for the first time. While the majority of students recognize the importance of financial education, our findings reveal notable gaps in their actual financial knowledge and behaviors, particularly in areas such as budgeting, saving, and managing digital financial tools.

The analysis indicates that certain demographic and academic variables, such as field of study, academic year, and employment status, significantly influence students' financial confidence and habits. For instance, students in business-related programs tend to demonstrate greater familiarity with financial concepts, while those in earlier academic years are often less confident and less consistent in managing their finances. Furthermore, students engaged in part-time work appear to have a more practical understanding of budgeting, though this does not always translate into better saving habits.

These insights align with previous research, such as Lusardi & Mitchell (2014) and Chen & Volpe (1998), which emphasize the long-term benefits of early financial education and highlight the widespread lack of essential financial skills among students.

Based on these findings, we recommend the implementation of mandatory financial literacy workshops tailored to different academic disciplines and year levels. In addition, universities should promote the responsible use of digital banking tools and applications by integrating them into student orientation programs or through targeted campaigns. These initiatives can play a pivotal role in bridging the gap between awareness and practice, ultimately fostering more financially resilient graduates.

■ 6. LIMITATIONS OF THE STUDY (Separator - 7)

- The use of a non-random sampling method limits the generalization of the findings to the broader student population.
- Reliance on self-reported data may introduce biases, such as social desirability or inaccurate recall.
- The brevity of the survey restricted the depth of questions related to investment knowledge, potentially oversimplifying students' understanding in this area.
- The cross-sectional design captures a snapshot in time and cannot assess changes in financial behavior or knowledge over time.
- The study primarily focused on university students, excluding other young adult populations who may have different financial experiences.
- External factors such as family background, cultural influences, or current economic conditions were not considered, which may significantly impact financial attitudes and behaviors.
- Limited demographic variables (e.g., parental income, employment status) may have restricted the scope of the analysis and its ability to identify deeper trends

■ 7. REFERENCES (APA style) (Separator - 8)

(Mandell, 2009)

(Mitchell, 2014)

(Volpe, 1998)

Bibliography

Mandell. (2009). The Impact of Financial Education in High School and College on Financial Literacy and Subsequent Financial Decision Making.

Mitchell, L. &. (2014). The Economic Importance of Financial Literacy: Theory and Evidence.

Volpe, C. &. (1998). An Analysis of Personal Financial Literacy Among College Students.

■ 8. APPENDIX (Separator - 9)

● Sample questionnaire

https://docs.google.com/forms/d/e/1FAIpQLSfM3k_2_5jElJPsgddWv2i--VXjz7YM38KYtJrmA4BduO1tOA/viewform

What is your age? *

- ☐ Below 18
- ☐ 18-20
- ☐ 20-25
- ☐ Above 25

What is your gender? *

- ☐ Male
- ☐ Female

What is your degree program? *

- ☐ School Of Business
- ☐ School of Computer Science
- ☐ School of Engineering
- ☐ School of Medicine
- ☐ School of Arts

What year of study are you in? *

- ☐ 1st year
- ☐ 2nd year
- ☐ 3rd year
- ☐ 4th year

Are you currently employed part-time? *

☐ Yes

☐ No

On a scale of 1–5, how confident are you in managing your personal finances ? *

1

2

3

4

5

Not confident at all

☐☐☐☐☐

Very confident

Do you know how to create and follow a personal budget? *

☐ Yes

☐ No

☐ Somewhat

Do you understand how credit cards and interest rates work? *

☐ Yes

☐ No

☐ Somewhat

Have you ever taken a course or workshop related to financial literacy? *

☐ Yes

☐ No

How often do you track your monthly expenses? *

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Always

What percentage of your monthly allowance/income do you usually spend? *

- ☐ 0–20%
- ☐ 21–40%
- ☐ 41–60%
- ☐ 61–80%
- ☐ 81–100%

What do you spend the most money on? *

- ☐ Food
- ☐ Transport
- ☐ Shopping (Clothes, Accessories)
- ☐ Entertainment (Movies, Games, etc.)
- ☐ Tuition/Fees
- ☐ Other: _____

Do you save a part of your income/allowance every month? *

- ☐ Yes
- ☐ No

What percentage of your income/allowance do you usually save? *

- ☐ 0%
- ☐ 1–10%
- ☐ 11–20%
- ☐ 21–30%
- ☐ 31% or more

Where do you usually save your money? *

- ☐ Bank Account
- ☐ Mobile Wallet (EasyPaisa, JazzCash, etc.)
- ☐ Cash at Home
- ☐ Investments (Stocks, Mutual Funds, etc.)
- ☐ Other: _____

What is your main reason for saving money? *

- ☐ Emergency Fund
- ☐ Education
- ☐ Travel
- ☐ Future Investments
- ☐ Other: _____

Do you believe financial literacy is important for young adults? *

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

Would you be interested in attending a financial literacy workshop? *

- ☐ Yes
- ☐ No
- ☐ Maybe

In your opinion, should universities offer mandatory financial literacy courses? *

- ☐ Yes
- ☐ No
- ☐ Maybe

Rank the following factors based on how much they influence your spending decisions *

	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Peer Pressure	☐	☐	☐	☐	☐
Advertisements	☐	☐	☐	☐	☐
Family Advice	☐	☐	☐	☐	☐
Personal Needs	☐	☐	☐	☐	☐
Personal Wants	☐	☐	☐	☐	☐

Submit

Clear form